

ARPIT THOOL

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Education

Virginia Tech

PhD in Computer Science

August 2021 – January 2026

Blacksburg, VA

Virginia Tech

Master's in Computer Science

August 2021 – May 2024

Blacksburg, VA

University of Pune

Bachelors of Engineering in Computer Science

August 2016 – May 2020

Pune, India

Experience

Adobe

Research Scientist

February 2026 – Present

Seattle, WA

- Advancing AI-powered creative tools through applied research in video, depth estimation, and generative media at Adobe's Research Engineering and Design lab.

Virginia Tech

Graduate Teaching Assistant

August 2025 – December 2025

Blacksburg, VA

- Assisted in teaching CS5244, focusing on technologies for modern data-centric web applications.

Adobe

Research Engineer Intern

May 2025 – August 2025

Seattle, WA

- Contributing to Firefly's Eval-Hub, an a platform for evaluating and benchmarking AI models used across product teams.
- Migrated the frontend and backend services infrastructure from East Coast to West Coast, reducing latency and improving performance for the primary user base.
- Optimized application performance by implementing Redis caching system for critical API endpoints, reducing leaderboard and evaluation page load times by 90%+ through strategic caching, automatic cache invalidation on data mutations, and seamless integration with existing FastAPI backend and MongoDB database infrastructure
- Collaborated with the Eval Strategy team to design the system architecture and UX for study-based validation, implementing efficient MongoDB queries using indexes, robust API endpoints, and backend services to embed gold-standard questions in studies.

Virginia Tech

Software Engineer

August 2021 – May 2025

Blacksburg, VA

- Researched, designed, implemented, tested and maintained software applications, writing clean code with version control, and appropriate unit and functional tests.
- Developed a Java software system to communicate effectively with OWASP ZAP, conducting thorough DAST scans on web applications, leveraging API calls, processing JSON data, and generating vulnerability reports.
- Deployed an AWS distributed load testing instance through a configured template and designed intricate JMeter scripts to conduct load testing of the login service effectively; done every semester before the first day of classes to ensure a smooth login experience.
- Designed & implemented a Python(Flask) web app with CAS SSO login feature to store and manage files on an on-premise RedHat server. This app is used by internal users over the VPN.
- Designed and implemented a Python-Flask middleware security solution bridging Splunk and Grouper, automating fraud detection workflows to process real-time alerts and manage user access controls. This system reduced fraud recognition time by 90% compared to manual processes, enhancing security and operational efficiency.

Adobe

Research Engineer Intern

May 2024 – Aug 2024

Seattle, WA

- Worked in the Research Engineering and Design lab on the [Stardust](#), an object-based image editor powered by AI and Adobe Firefly. I built the image analysis pipeline using a node-based graph framework for data processing and building interactive workflows.
- The new pipeline resulted in better performance, flexibility, and deployment, allowing researchers to rapidly iterate and experiment to build new 2D and 3D features.
- Implemented a 3D feature to move a selected object within a 2D image in a perspective-aware manner, resizing and reordering layers as needed.
- Wrote quality TypeScript code in the AdonisJS framework using the MVC design pattern. Undertook system design and UX decisions to optimize the system performance & scalability and enhance usability.

Lutron

Software Engineer Intern

May 2022 – August 2022

Coopersburg, PA

- Collaborated with tech lead, product owner and senior engineers to understand requirements, create UML diagrams and to create the documentation for the system architecture.
- Spearheaded design, development and deployment of a large scale Java Spring application used to migrate from manual management of a Cloud resource to Terraform, slashing manual process time by 50%.
- Wrote CI/CD pipeline script in Groovy to connect the BitBucket repository to the Terraform Provider which in turn performed required changes on Lutron's Artifactory instance.
- Extended functionality of Lutron's System Diagnostics app to enable debug logging for a connected system using TypeScript and React.

Acquia

Associate Software Engineer (Full Stack)

September 2020 – June 2021

Pune, India

- As a part of the Marketing Cloud team, I worked on the Campaign Factory and Campaign Studio products.
- Collaborated with the product owner and senior engineers to participate in code reviews and discuss innovative solutions to add new functionality and increase system efficiency.
- Modeled and implemented an efficient solution to synchronize the creation, modification and deletion of an asset in the parent template to children templates, saving 2000 clients 90% of their time spent on duplicating templates.
- Worked closely with the QA and the UX teams to ensure the features are built according to the requirements of the customers using VueJS, Bootstrap, HTML and CSS.
- Mentored 3 new college graduates joining Acquia. Conducting weekly progress meetings and provided guidance on revising and focusing on key concepts and technologies for a seamless transition to their respective projects.

Projects

Multi-GPU LLM Fine-Tuning Pipeline | [Python](#), [PyTorch](#), [Hugging Face](#), [Accelerate](#), [PEFT](#), [BitsAndBytes](#)

October 2025

- Developed a distributed fine-tuning pipeline for large-context LLaMA/Qwen models using 4-bit QLoRA, running on multi-GPU HPC clusters with gradient checkpointing and memory-efficient adapters.
- Integrated accelerate for distributed training, early stopping, and BERT & ROUGE evaluation; visualized loss and metric curves across epochs.

AI Agent — [Hugging Face Space Link](#) | [Python](#), [MCP](#), [SmolAgent](#)

September 2025

- Deployed on Hugging Face : A text-based LLM chat app using the lightweight Qwen model, with an agentic backend that can fetch local time by location, and search the web based on requirements via tool integrations.
- Built the agent with Smolagents (CodeAgent), leveraging its MCPClient to connect to external MCP servers and integrate their tool collections into the workflow.

SafeAIMerge — [GitHub Link](#) | [Python](#), [GitHub-Actions](#), [Docker](#), [ZAP](#), [LLM-API](#)

September 2025

- Developed an automated DevSecOps tool that integrates ZAP and OpenAI within GitHub Actions to perform dynamic web app security scanning, generate natural-language summaries of findings, and post actionable feedback directly on pull requests.
- Implemented a configurable pipeline-gating framework that classifies and enforces risk levels via YAML settings, enabling automated build failures for critical alerts and improving developer-centric security feedback loops.

Publications

- **Integrating Log-Based Security Analytics in Agile Workflows: A Real-World Experience Report** — [HCSE 2026](#)
International IFIP Working Conference on Human-Centered Software Engineering
- **SafeAIMerge: A Tool for Integrating DAST and LLM-Generated Security Feedback into GitHub Actions Workflows** — [SecDev 2026](#) ACM Secure Development Conference
- **Practitioner Perspectives of DAST Integration in Agile Development Workflows: An Experience Report** — [FSE 2026](#)
34th ACM International Conference on the Foundations of Software Engineering
- **Harnessing the Power of LLMs to Simplify Security: LLM Summarization for Human-Centric DAST Reports** — [VLHCC 2024](#) IEEE Symposium on Visual Languages and Human-Centric Computing
- **Securing Agile: Assessing the Impact of Security Activities on Agile Development** — [EASE 2024](#) 28th International Conference on Evaluation and Assessment in Software Engineering

Technical Skills

Programming Languages: Java, Python, Go, C#, C++, Javascript, TypeScript, PHP
Database Technologies: MySQL, Redis, NoSQL, MongoDB
Frameworks: Spring, Spring Boot, Flask, FastAPI, Django, React.js, JavaServer Faces, Angular, Vue.js, Node.js
Other Technologies: MCP, CodeAgent, Terraform, AWS, GCP, Docker, Kubernetes K8, REST APIs, Git, GitHub, GitLab, HTTP, JSON, TDD, XML, Linux, Ubuntu, shell script, Jenkins, HTML5, Groovy, DAST
Industry Skills: Backend & DevOps, Test Driven Development, SOLID Principles, Agile, Object Oriented Programming, CI/CD, Data Structures and Algorithms, Microservices